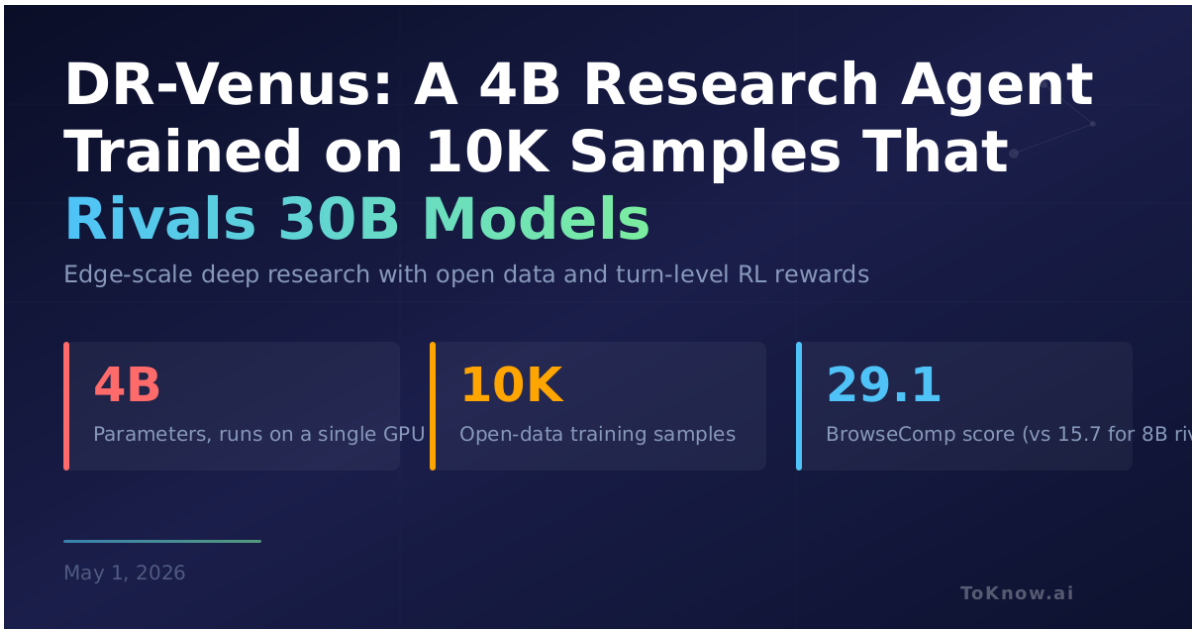
DR-Venus: A 4B Research Agent Trained on 10K Samples That Rivals Models 8x Its Size

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Read at ToKnow.ai



Inclusion AI released DR-Venus, a 4B-parameter deep research agent built entirely on open data that outperforms every prior agent under 9B parameters on multiple research benchmarks. The model is built on a Qwen3-4B backbone and trained in two stages. First, agentic supervised fine-tuning on cleaned trajectories from REDSearcher: 10,001 raw research traces are pruned to 9,365 correct ones, then upweighted via turn-aware resampling into 18,745 training instances.

Second, reinforcement learning using IGPO with turn-level rewards. Instead of a single score at the end of a multi-step research session, each turn gets two signals: an information gain reward (did the agent actually learn something new from that search or page visit?) and a format-aware penalty (is it maintaining structured notes and proper citations?). On BrowseComp, DR-Venus-4B-RL scores 29.1, nearly doubling the 15.7 from WebExplorer-8B-RL, a model twice its size. It handles up to 200 tool-call steps with 256K context at inference.

The practical upside: a capable research agent that fits on a single consumer GPU. At 4B parameters with GGUF quantized versions available, anyone with a laptop GPU can run an agent that autonomously searches the web, reads pages, and synthesizes answers. The 10K open-data requirement means any lab can reproduce and extend this without proprietary access.

The same Inclusion AI team behind LLaDA2.0-Uni is building a pattern: high-impact open-source releases that compress frontier capabilities into reproducible, small-scale packages.

Sources:

- DR-Venus: Towards Frontier Edge-Scale Deep Research Agents with Only 10K Open Data (arXiv)
- DR-Venus GitHub Repository
- DR-Venus-4B-RL Model (Hugging Face)
- REDSearcher Open Trajectory Data

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